

STUDYFORGE EDGE

Private On-Device AI Study Copilot for Snapdragon PCs

1. The Problem

- Students move lecture notes into cloud AI tools.
- Study workflows are fragmented.
- Privacy and connectivity are concerns.

2. The Solution

- Upload notes once.
- Generate an exam-focused summary, quiz, revision plan and grounded Q&A.;
- Keep the core AI workflow local.

3. Snapdragon AI

- Qualcomm AI Hub models + GenieX.
- Target Snapdragon X-series NPU execution.
- Local OpenAI-compatible API keeps the application layer simple.

4. Architecture

- Student → UI → PDF extraction → GenieX → Snapdragon NPU → outputs.
- MVP model: Qwen3-4B-Instruct-2507.

5. Innovation

- Workflow-driven rather than generic chat.
- Understand → Practice → Revise.
- Privacy is a product feature.

6. MVP Demo

- Upload PDF.
- Generate summary.
- Generate quiz.
- Generate 3-day plan.
- Ask a grounded question.

7. Technical Stack

- Python, Streamlit, PyMuPDF, GenieX, Qualcomm AI Hub model.

8. Evaluation

- Latency, memory, NPU utilization, output quality, offline completion.
- Only report measured results.

9. Roadmap

- Vision-language diagrams.
- Speech input.
- Adaptive revision.
- Packaged Windows app and NPU dashboard.

10. Closing

- Private. Local. Useful.
- StudyForge Edge demonstrates a practical on-device AI workflow for Snapdragon PCs.